

GCSE algebra - Set 1 - Solutions

Specification M4.1-M4.8, M4.18, M4.19.

Answer key

Question	Answer	Topic
1	E	GCSE algebra
2	A	GCSE algebra
3	A	GCSE algebra
4	C	GCSE algebra
5	A	GCSE algebra
6	D	GCSE algebra
7	C	GCSE algebra
8	E	GCSE algebra
9	F	GCSE algebra
10	F	GCSE algebra
11	C	GCSE algebra
12	C	GCSE algebra

Worked solutions

Question 1 ALG-01 · GCSE algebra · spec M4.7

Make x the subject of $y = \frac{2x+3}{x-1}$.

Answer: E

$$y(x-1) = 2x+3 \Rightarrow xy-2x = y+3 \Rightarrow x(y-2) = y+3 \Rightarrow x = \frac{y+3}{y-2}.$$

Question 2 ALG-02 · GCSE algebra · spec M4.19

Find the n th term of the quadratic sequence 3, 8, 15, 24, 35, ...

Answer: A

Second differences are 2, so the n th term is $n^2 + bn + c$: $n = 1$ gives $1 + b + c = 3$ and $n = 2$ gives $4 + 2b + c = 8$. Subtracting, $3 + b = 5$, so $b = 2$ and $c = 0$: $n^2 + 2n = n(n+2)$.

Question 3 ALG-04 · GCSE algebra · spec M4.8

Which of the following is an identity (true for all values of x)?

Answer: A

$$(x+3)^2 - (x-3)^2 = (x^2 + 6x + 9) - (x^2 - 6x + 9) = 12x \text{ for all } x.$$

Question 4 ALG-05 · GCSE algebra · spec M4.18

A car accelerates uniformly from rest to 20 m/s in 8 seconds, travels at 20 m/s for 12 seconds, then decelerates uniformly to rest in 10 seconds. Find the total distance travelled.

Answer: C

$$\text{Area under the speed-time graph: } \frac{1}{2} \cdot 8 \cdot 20 + 12 \cdot 20 + \frac{1}{2} \cdot 10 \cdot 20 = 80 + 240 + 100 = 420 \text{ m.}$$

Question 5 ALG-06 · GCSE algebra · spec M4.3

Given that $v^2 = u^2 + 2as$, find s when $v = 13$, $u = 5$ and $a = 4$.

Answer: A

$$169 = 25 + 8s \Rightarrow s = 18.$$

Question 6 ALG-07 · GCSE algebra · spec M4.6

Simplify $\frac{3}{x+2} - \frac{2}{x-1}$.

Answer: D

$$\frac{3(x-1) - 2(x+2)}{(x+2)(x-1)} = \frac{x-7}{(x+2)(x-1)}.$$

Question 7 ALG-09 · GCSE algebra · spec M4.7

Given that $T = 2\pi\sqrt{\frac{l}{g}}$, express l in terms of T and g .

Answer: C

$$\frac{T}{2\pi} = \sqrt{\frac{l}{g}} \Rightarrow \frac{T^2}{4\pi^2} = \frac{l}{g} \Rightarrow l = \frac{gT^2}{4\pi^2}.$$

Question 8 ALG-10 · GCSE algebra · spec M4.18

A ball is thrown vertically upwards and its height after t seconds is $h = 20t - 5t^2$ metres. For how long is the ball at least 15 m above the ground?

Answer: E

$$20t - 5t^2 \geq 15 \Rightarrow t^2 - 4t + 3 \leq 0 \Rightarrow 1 \leq t \leq 3: \text{two seconds.}$$

Question 9 ALG-11 · GCSE algebra · spec M4.4

Expand and simplify $(x + 2)(x - 1)(x + 3)$.

Answer: F

$$(x^2 + x - 2)(x + 3) = x^3 + 3x^2 + x^2 + 3x - 2x - 6 = x^3 + 4x^2 + x - 6.$$

Question 10 ALG-12 · GCSE algebra · spec M4.19

A sequence has first term 2 and each subsequent term is obtained by squaring the previous term and subtracting 2. What is the 50th term?

Answer: F

$$2^2 - 2 = 2: \text{every term is 2.}$$

Question 11 ALG-13 · GCSE algebra · spec M4.5

Factorise $6x^2 - 7x - 20$.

Answer: C

$$(3x + 4)(2x - 5) = 6x^2 - 15x + 8x - 20 = 6x^2 - 7x - 20.$$

Question 12 ALG-14 · GCSE algebra · spec M4.16

The sum of two numbers is 12 and the difference of their squares is 48. Find the larger number.

Answer: C

$$x^2 - y^2 = (x - y)(x + y) = 12(x - y) = 48 \Rightarrow x - y = 4. \text{ So } x = 8, y = 4.$$